

and Clothing in Hong Kong, it is a highly-stretchable fabric formulation that can withstand multiple machine washes. The material has multiple filaments of pre-stretched elastic yarn and polyurethane-coated copper fibres. These are combined to form a thin material that can have microchips and components fixed on it, whose electrical connections are maintained even when stretched.

An interesting example of these materials being used is in a policeman's bullet-proof Kevlar vest. If the policeman is shot, then the material senses the bullet's impact and calls for back-up automatically. This allows the policeman to focus on the situation in front of him.

Another example of electronics-infused textiles started right here in India. Nihal Kashinath, founder, IoT-BLR, one of the largest IoT-focused meet-up groups in the world, says, "At IoTBLR, we prototyped a program-mable shirt—a shirt on which you can load apps, such as music controller, slide-show controller, remote-camera controller, movement tracker and panic button. The project was spun-off as a separate company, ThingsEvolue, where specific use-case shirts and other apparel are being developed."

With great features comes great power consumption

The power problem is certainly not an easy one to solve, particularly in the wearable segment. "Today's technologies are allowing engineers to harvest and manage low-power (nano) from a variety of sources, including solar cells, thermoelectric cells, electromagnetic and vibration energies. These technologies are providing designers with more options to charge batteries while the device is in use," says Sachidananda Karanth, senior technical architect, Mistral Solutions.

Energy harvesting from movement, heat, light and even ambient radiation is being explored for wearables. Nihal explains how the selection of energy-harvesting method depends heavily on use-case (if it is a sleep tracker,

How nanotechnology could affect wearable electronics

Innovators at NASA Glenn Research Center have developed a suite of nanotechnologies that offer the potential to enhance or replace silicon based technology for high-speed, high-computing power devices that operate at low voltage and current levels. These nanotechnologies are used to make polymer nanowires, carbon nanotubes (CNTs) and other nano-ionic structures, which produce a variety of nano-scale electronic switches, which are the basic building blocks for relatively inexpensive, low-voltage and high-speed logic circuits. The nanotechnologies are more durable, with longer lasting circuits, and enable computational capabilities at specific locations.

These circuits can be used to make sensing systems, wireless and embedded communications, large-area flexible electronic displays, active matrix light emitting diode (LED) displays, along with antennae and radiofrequency identification (RFID) devices, such as smart keys and smart cards. These could also find applications in electronic-circuitry-protection elements.

—James Colby, manager, business and technology development semiconductor business unit at Littelfuse

kinetic-energy harvesting would be a poor choice) and power requirements of the device's features (if your wearable camera needs to be used for advanced image processing, energy harvesting will not be of much help).

Thin-film-battery technology is already being implemented in electronics, like Biostamp from MC10, that has partnered with Braemar Energy Ventures for its solutions. One example is Solicore, which provides flexible thin-film lithium-polymer batteries that are less than half a millimetre thick and yet hold up to 25mAh while delivering three volts. These allow design engineers to disregard bulky conventional battery designs while developing the product.

New wearable designs must support an increase in functional integration, as well as offer a reduction in overall power consumption, cost and physical footprint. "These cost-sensitive embedded designs require a move away from the traditional arms race of MIPS, bytes and megahertz, and a paradigm shift into the era of function enablement," explains Marten Smith, marketing manager, Medical Products Group, Microchip Technology Inc.

Another technology available is for those devices whose design enables these to gain from harvesting energy from the environment, like sunlight. What is required is a high-efficiency thin-film-photovoltaic technology. Organic photovoltaic seem to be one potential candidate here. One com-

Bend, stretch, flex and back. These are words you usually hear in those early morning fitness programmes on TV, but these now apply to circuits too

pany, named Solarmer Energy, has been successfully working on organic photovoltaic technology that could be used to make power-generating fabrics. Other firms in this segment are Eight19, Heliateg and Dyesol.

Korea Advanced Institute of Science and Technology (KAIST) has a team working on developing a flexible glass fabric based thermoelectric (TE) power generator that produces electricity from the heat generated by a human body. Their journal states that they used a screen-printing technique to build the generator, which can be used to power self-sustaining devices. "With this technique, it is possible to make the device thin ($\sim 500\mu\text{m}$), light-weight ($\sim 0.13\text{g cm}^{-2}$) and flexible," light enough for wearable devices.

At CES this year, Intel's CEO, Brian Krzanich, plucked a button from his blazer and explained how that is Intel's new Quark SE SoC-powered device named Curie, their latest development module targeting the wearable market.

Deepak adds that a California based start-up, Imprint Energy, is developing flexible and rechargeable bat-